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Onderwerp

Mardie salt ponds: idealized morphodynamic model

1 Introduction

The future development of solar sulphate and salt production facilities along the Pilbara coast by Mardie Minerals Pty Ltd fuels the need for better understanding the fate intertidal benthic communities and habitats along this stretch of the Western Australia coastline. This has resulted in a research project 'Quantifying the effects of climate change on intertidal habitats of the west Pilbara coast' led by the University of Western Australia, aiming to advance our understanding of the intertidal and supratidal flats in the context of climate change, especially sea level rise (SLR). Part of the work to be carried out is the development of numerical models that are able to simulate the morphological development of the intertidal and supratidal flats under the influence of SLR but influenced by (1) the algal mats typical for the Pilbara coast; (2) salt crust formation; and (3) mangroves.

The final goal of this project is to develop model applications able to meet the above requirements. Probably the two most challenging aspects of this modelling effort are that (1) morphological changes resulting from SLR are subtle, requiring a model that is in equilibrium (i.e. not in a process of morphological spin-up). Morphological models based on realistic bathymetries are usually not in equilibrium, but morphologically adapting until the computed morphology is in equilibrium with the modelled boundary conditions, model settings and modelled processes. Changes resulting from spin-up are often stronger than changes resulting from SLR, complicating quantifying the contribution of SLR.

(2) the effect of the intertidal communities typical for the Pilbara coast and the formation of salt crust on the stability of the sediment bed are not implemented in commonly used mainstream morphological models.

Morphological equilibrium is much more easily achieved using geometrically schematized models (hereafter referred to as idealized models). Such models are initiated with a simplified bathymetry (typically width-averaged with a constant, linearly sloping or convex-concave-upward longitudinal slope), and attain conditions approaching morphological equilibrium. Subsequently implementing features with a subtle to moderate effect on the bathymetry (SLR, algal mats, salt crusts) and comparing these with the original model scenario allows quantification of the impact of SLR, algal mats and salt crusts.

The role of Deltares in the Pilbara project is to (1) support the development of models (idealized as well as complex) and (2) implement processes in Delft3D (such as salt crusts and the algal mats). In order to numerically experiment with the effect of SLR and new model features, Deltares has developed an idealized model. The current note reports on the setup and main findings of an idealized model resembling part of the Pilbara coast. For this purpose we have chosen a tidal creek system in the Giralia section for two reasons: (1) it is relatively confined (limited connection with other creeks, making it more straightforward to model and interpret) and (2) hydrodynamic data has been collected as part of the project (allowing more advanced

calibration in future stages of the project). The model is developed in Delft3D FM (and not Delft3D-4) because of the advanced functionalities available in BMI (Basic Model Interface) which is part of Delft3D-FM but not in Delft3D-4.

In this first model development stage we evaluate the model sensitivity to erosion using readily available functionalities (sand-mud interaction settings as well as erosion properties). Implementation and testing of new model functionalities influencing the bed erosion (algal mats, salt crusts) will be part of future efforts.

2 Model set up

A Delft3D FM morphodynamic model is set up to obtain a bathymetry in dynamic equilibrium. The settings are based on the model used by Colina Alonso et al. (2023). The characteristics of a tidal channel system of the Giralia area are taken into account. This includes a large main channel with a width that decreases from 263 meter at its seaward end up to ~14 meter (although eventually finer in secondary branches) at the landward end, see Figure 1.

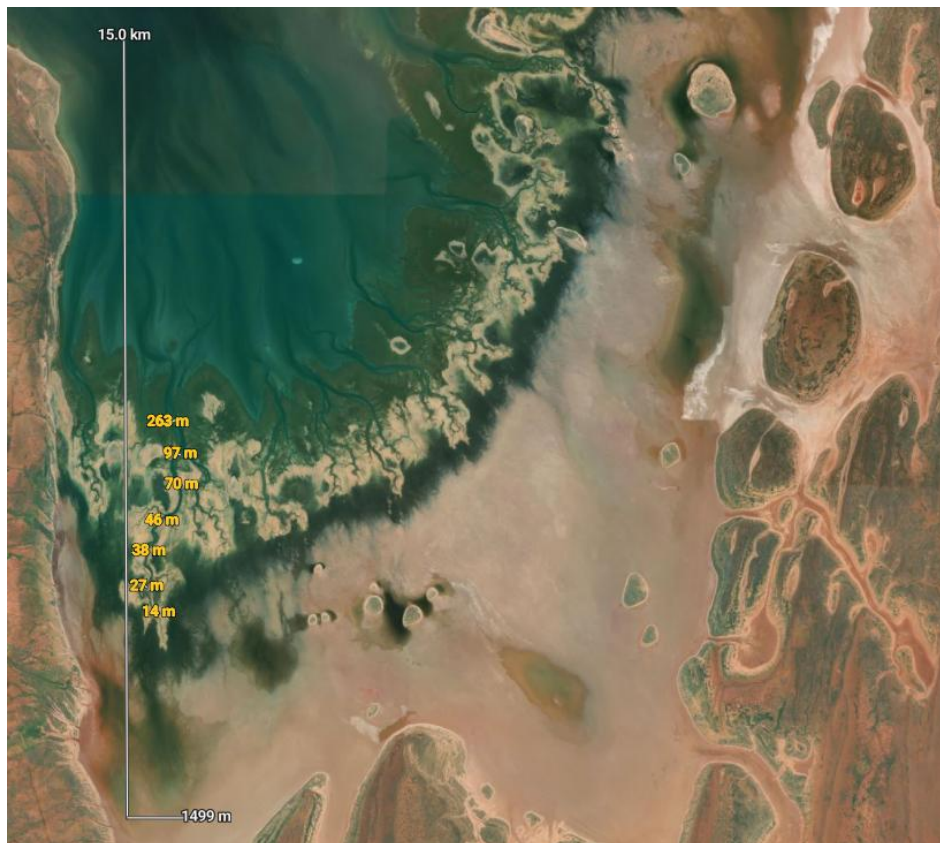


Figure 1 Tidal creek in the Giralia area, south Exmouth bay showing dimensions of the idealized model (white) and the actual width of the tidal creek (yellow). Image and measurements from TomTom (2024).

2.1 Grid

To capture the extent of one of the main tidal creeks in the Giralia area, a rectangular computational grid is constructed with a length of 15 km (longshore) and a width of 1.5 km (cross-shore). The grid has a resolution of $dx=dy=15$ meter, which ensures the main channel is represented by at least one cell (at the landward end) and more than 15 cells at the seaward end. For the secondary branches, characterized by channel widths in the order of 5 meter, a finer resolution might be desirable, but this will increase the computational time significantly

$((15/5)^2 = 3^2 = 9$ times as many cells, combined with a timestep which may be 3 times smaller up to 27 times slower). Such adaptation may be explored in a later stage. The grid is constructed with the Deltares python toolbox `dfm_tools` (Veenstra, 2024).

2.2 Bathymetry

The initial idealized bathymetry is assumed longshore uniform with a convex profile shape that is associated with muddy coastlines. The shape is obtained by fitting an exponential function to bathymetrical data obtained from GEBCO (2023) for the selected domain in the Giralia region (see Appendix 5A) and averaged in the longshore direction, see blue line in Figure 2. The following function is used to describe the cross-shore bathymetry:

$$h(y) = a * e^{(b-c*y)} + d.$$

Here, $h(y)$ is the bed level at a certain longshore distance y , and a, b, c and d are parameters obtained by a non-linear least squares fitting procedure, which yields the values $a = -0.730, b = 0.0351, c = -1.65 * 10^{-4}, d = 3.50$. See the orange line in Figure 2 for the obtained idealized cross-shore elevation profile. The vertical datum of the bathymetry data is 1.45 meter above Mean Sea Level.

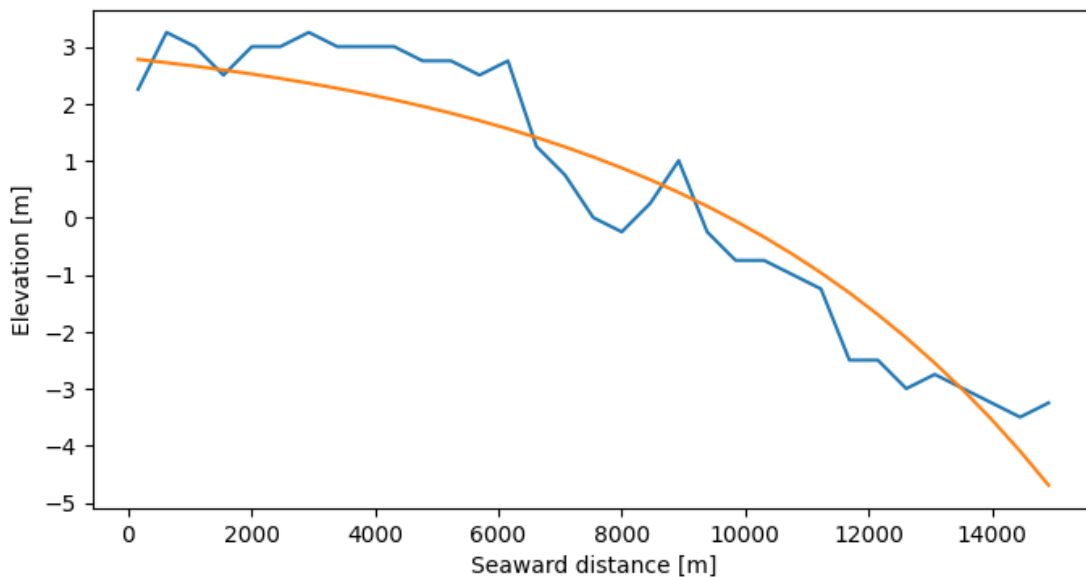


Figure 2 Convex cross-shore bathymetry profile for the idealized model domain (orange line) obtained by fitting Equation 1 through the longshore-averaged GEBCO bathymetry of the selected area (blue line).

2.3 Hydrodynamic forcing: tides

Tidal forcing is imposed on the seaward end of the domain. Tidal components are obtained from the TPX08 global tide database (Egbert & Erofeeva, 2002, updated) for the location. An inverse distance transformation is applied between the TPX08 grid cells which are closest to the selected point. The obtained tidal amplitude and phases are found in Tabel 1. The constituents with an amplitude smaller than 0.1 m are considered insignificant and therefore neglected, leaving M2, S2, K1 and O1. Tidal asymmetry therefore results from (1) the interaction between the K1, O1 and M2 constituents (Hoitink et al, 2003) and (2) the overtides generated by friction within the model (mostly M4). The difference between the bathymetry datum and mean sea level (1.45 m) is prescribed as an A0 component.

Tabel 1 Tidal components as obtained from TPX08 for the selected site (marked by the star in the right panel).

Name	Amp. [m]	Phase [deg]
S2	0.346	-179.7
K1	0.2048	-158.8
K2	0.07901	179.3
M2	0.9387	109.3
M4	0.006083	80.54
N2	0.09851	95.83
O1	0.1338	-170.5
P1	0.0642	-161.8



The morphological development of this sheltered intertidal area seems to be steered mainly by flow caused by tides and other phenomena that cause high water levels. The wave action seems limited, and is not incorporated in the model setup to save computational costs.

2.4 Sediment in the domain

Two sediment fractions are included: sand ($D = 250 \mu\text{m}$) and mud ($<63 \mu\text{m}$). Following Colina Alonso et al. (2023), the initial bed is composed of 20 layers of 1 meter with a sediment composition of 95% sand and 5% mud (mass-fractions). On top of that is an active transport layer of 0.1 meter with initially the same composition. The results are sensitive to the thickness of the active layer as the sediment composition influences erosion. For a very thin layer (order of millimeters), changes due to erosion and deposition will strongly influence the sediment composition and with that erosion on the next timestep. For very thick layers (order of meters), the sediment composition hardly changes with erosion and deposition. On the northern boundary ($y = 15000$ meters) a mud concentration of 5 mg/l is imposed.

2.5 Sand-mud interaction

Sand-mud erosion interaction is included using the van Ledden (2004) model. Sediment in the bed is regarded a mixture with a certain mud (clay + silt) content (p_{mud}). Non-cohesive transport is computed with the van Rijn (1993) formula. If the mud content exceeds a critical value ($p_{\text{mud,crit}}$) the mixture cohesive and erosion is computed by the Partheniades erosion formula. In this approach, the erodibility threshold (τ_e) of both non-cohesive and cohesive mixtures is modulated by the mud content. No roughness interaction effects are considered. For details on the transport formulations see van Ledden et al. (2004); Colina Alonso et. al (2023).

Tabel 2 Model parameter settings

Parameter	Value	Description
n	0.023	Manning roughness coefficient ($\text{s/m}^{1/3}$)
MorFac	50	Morphological scale factor (-)
D_{50}	250	Median grain diameter of sand (μm)
w_s	$2.5\text{e-}4$	Settling velocity of mud (m/s)
τ_e	0.125-2	Critical bed shear stress for erosion of mud (N/m^2)
M	$1\text{e-}4$	Erosion parameter (m/s)
$p_{\text{mud,crit}}$	0.3	Critical mud content for cohesive behavior (-)
β_m	1	Bed packing coefficient (-)
$c_{\text{mud,bound}}$	5	Suspended mud concentration at the boundary (mg/l)

2.6 Scenarios

The model is run for three hydrodynamic years and a morphological acceleration factor (MorFac) of 50, resulting in 150 years of morphological development. With the aim of first attaining morphological equilibrium, no sea-level rise is imposed. We investigate the model sensitivity to erosion by first evaluating the effect of sand-mud erosion interaction (using $\tau_e = 0.5 \text{ N/m}^2$). Thereafter, the critical bed shear stress for erosion of mud is varied between 0.125 N/m^2 (easily eroded) and 2.0 N/m^2 (consolidated mud). The results are evaluated for morphological evolution of the bed level, mud content in the channels and on the shoals or boundaries.

3 Results

3.1 Effect of sand-mud erosion interaction ($\tau_e = 0.5 \text{ N/m}^2$)

The morphological development without sand-mud erosion interaction is gradual in the simulated 150 morphological years. The width-averaged elevation shows erosion between +2 m (y ~ 4000 m) and 0 m (y ~ 10000 m) and accretion between 0 m (y ~ 10000) and -2.5 m (y ~ 13000) (Figure 3). One main channel has formed in the simulated period (Figure 4). Mud erodes only for $\tau > \tau_e = 0.5 \text{ N/m}^2$, conditions that do not occur often (see later, Section 3.2). Therefore, p_{mud} increases in the erosion zone as mud in the bed is immobile and sand is eroded, whereas p_{mud} decreases in the deposition zone as sand is deposited and mud immobile, see Figure 5 **Error! Reference source not found.** This leads to a stabilization of the eroding zone as the higher mud-content increases the critical bed shear stress for erosion of sand. When channels start forming, higher flow velocities occur in the deeper parts and will exceed the erosion threshold for mud (occasionally). This does not occur on the shallower parts adjacent to the channels and mud accumulates there.

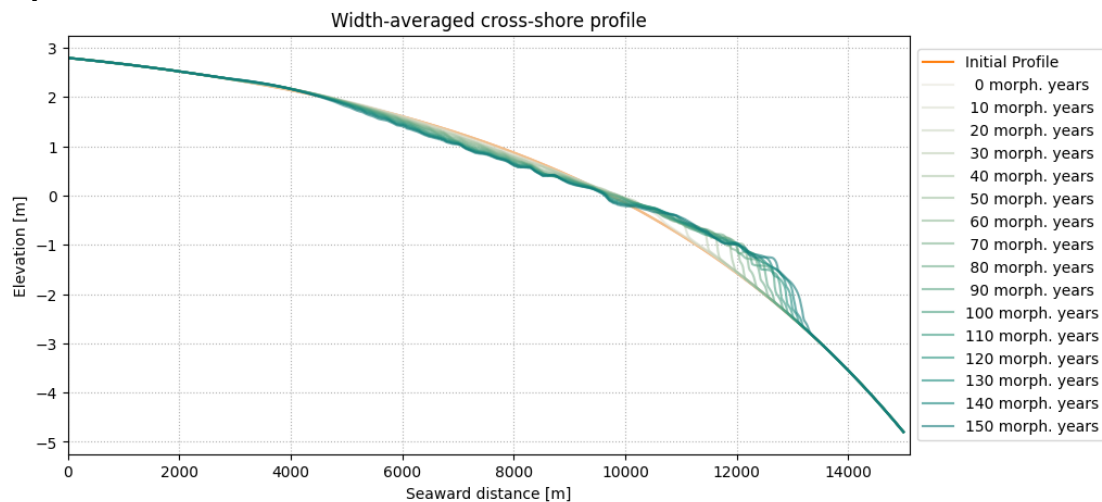


Figure 3 Width-averaged elevation vs seaward distance for $t_{mor} = 0, 10, 20, \dots, 150$ years without erosion interaction.

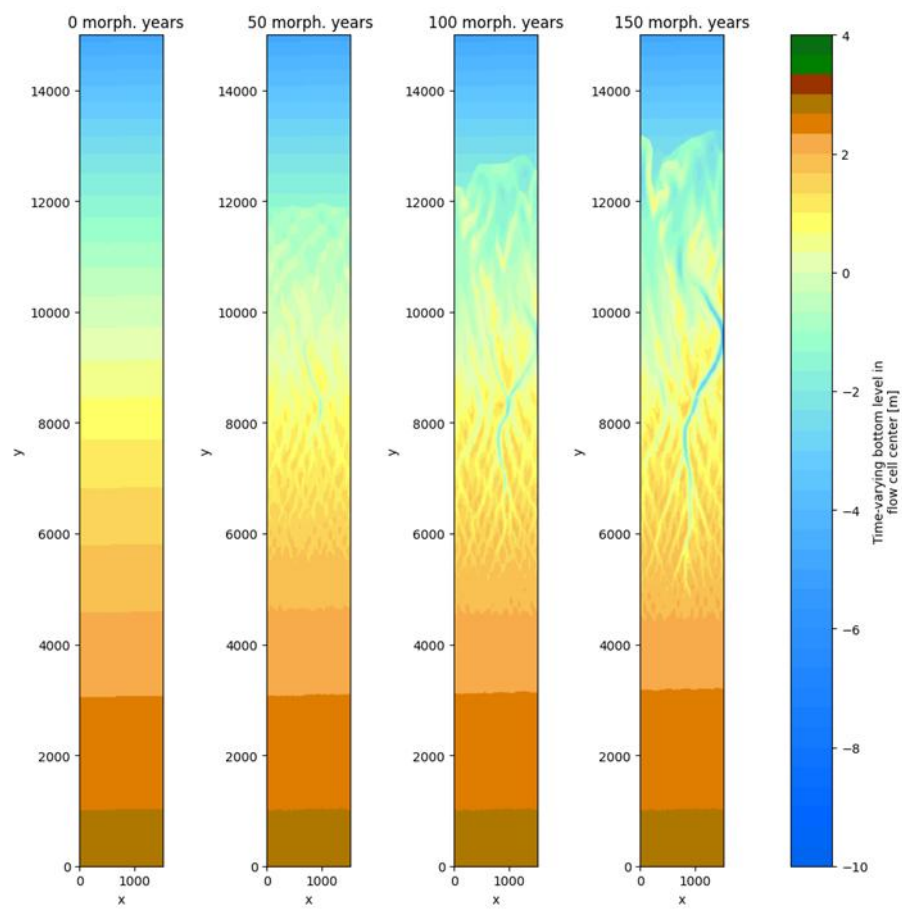


Figure 4 Bed level for $t_{mor} = 0, 50, 100, 150$ years without erosion interaction.

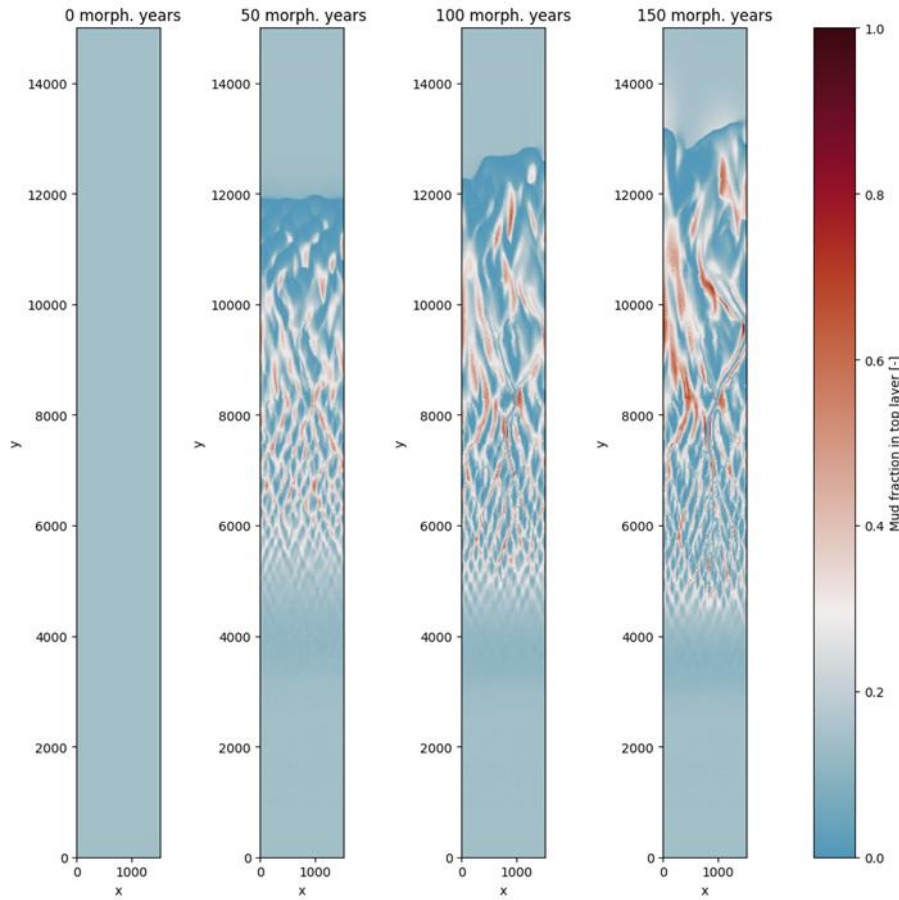


Figure 5 The fraction mud p_m in the active toplayer for $t_{mor} = 0, 50, 100, 150$ years without erosion interaction.

Significant increase in morphological development occurs when accounting for sand-mud erosion interaction. The erosion and deposition zones, while on similar location, are larger and show greater change compared to the simulation without interaction, see Figure 6. Large tidal channels have formed after 50 morphological years which develop until (and after) 150 morphological years, and are flanked by shallower shoals, see Figure 7. Significant bodies of sand are deposited towards the seaward end of the domain that raise the local bed level. The mud-content reflects the channel-shoal dynamics, with muddy zones (red in Figure 8) near the landward end and on the channel banks, where flow velocities are small. Furthermore, another muddy zone is at the offshore boundary where mud enters the domain and flow velocities are small. The mixture approach allows for erosion of both sand and mud, even when the critical erosion of (pure) mud $\tau_e = 0.5 \text{ N/m}^2$ is not exceeded. The result is a more dynamic channel-shoal system.

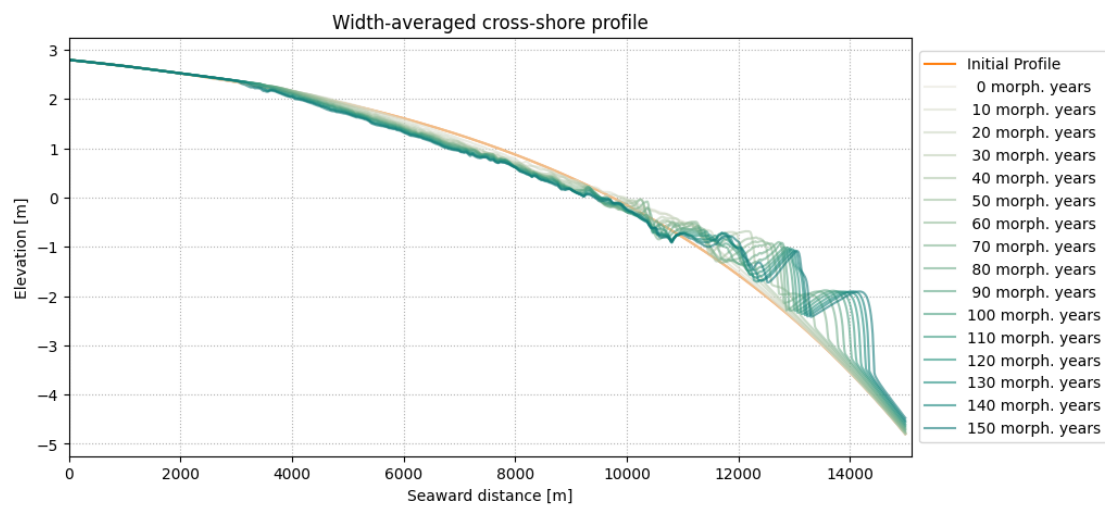


Figure 6 Width-averaged elevation vs seaward distance for $t_{mor} = 0, 10, 20, \dots, 150$ years with erosion interaction.

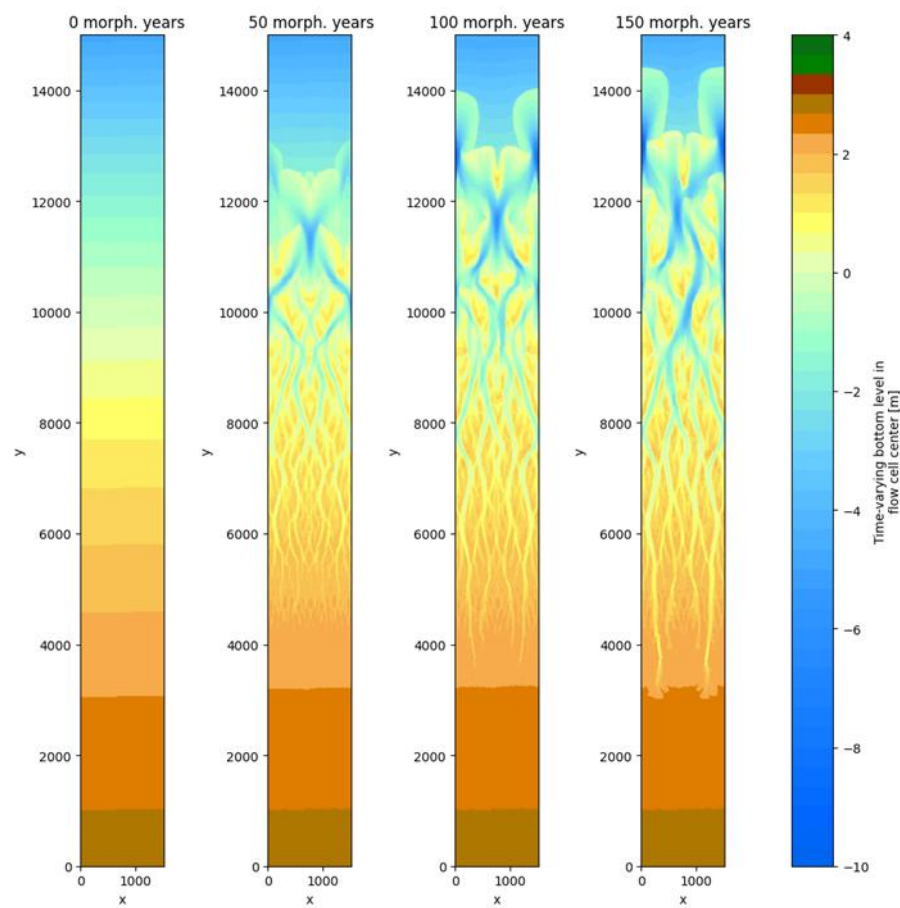


Figure 7 The bed level for $t_{mor} = 0, 50, 100, 150$ years with erosion interaction.

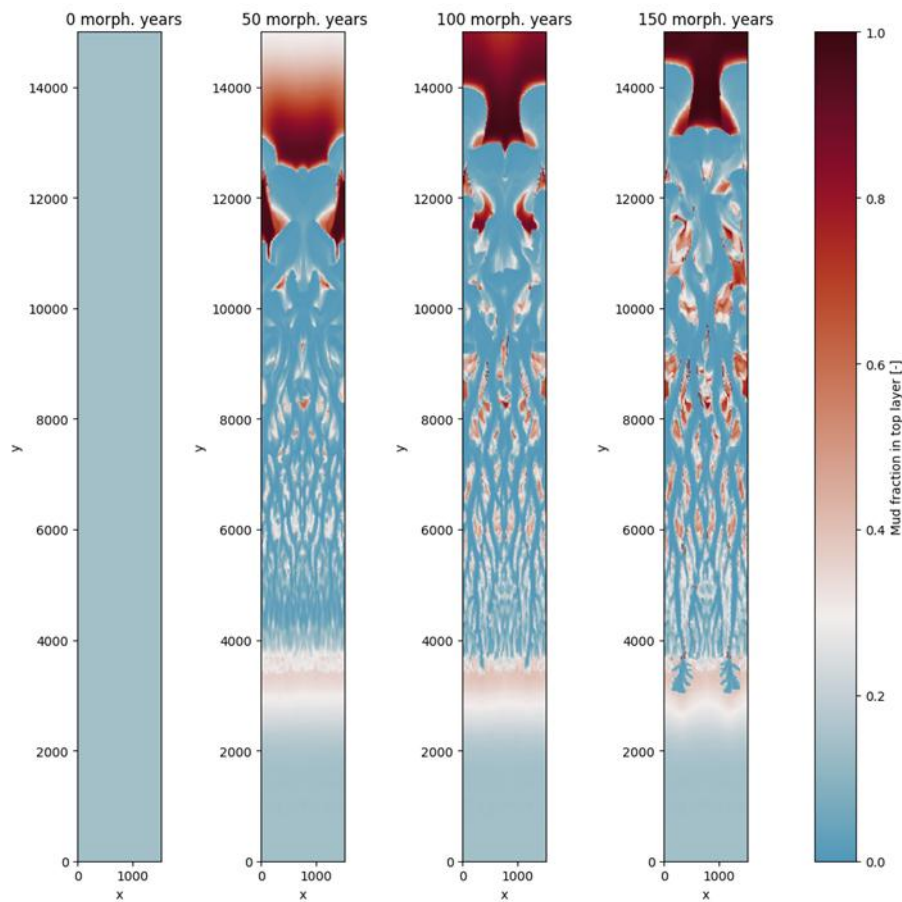


Figure 8 The mud fraction p_m in the active toplayer for $t_{mor} = 0, 50, 100, 150$ years without erosion interaction.

3.2 Bed shear stresses

Bed shear stresses vary between 0 and 1 N/m² and are highest in the channels. After nine months simulation time (37.5 morphological years), shear stresses of ~ 0.1 N/m² are exceeded fifty percent of the time. Magnitudes of ~ 0.2 N/m² are exceeded only ten percent of the time, and ~ 0.5 N/m² only one percent of the time.

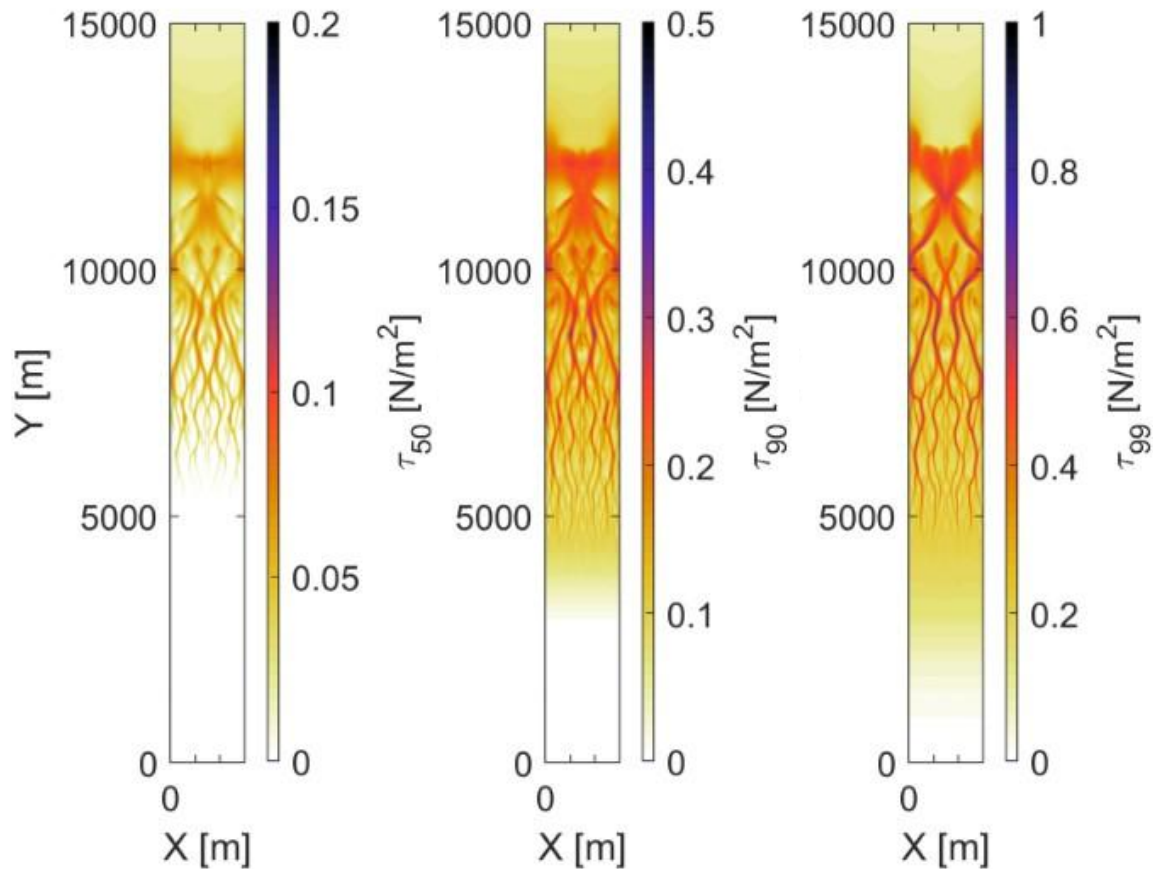


Figure 9 The magnitude of bed shear stresses after 9 months simulation time (37.5 morphological years) in the domain for the 50th percentile (left), 90th percentile (middle) and 99th percentile (right).

3.3 Sensitivity to critical bed shear stress for erosion of mud

Colina Alonso et al. (2023) show the sensitivity of channel-shoal patterns to the critical erosion of mud τ_e when applying sand-mud erosion interaction. Due to the absence of waves and generally more milder conditions (lower bed-shear stresses), less profound differences in channel-shoal patterns are found when varying the critical erosion of mud. Channel size and patterns are comparable for $\tau_e = 0.125, 0.25, 0.5, 0.75, 1, 1.5, 2$ N/m² as transport is mainly non-cohesive. More easily erodible mud ($\tau_e = 0.125, 0.25$) results in channels that reach less far landward, see Figure 10. The most erodible mud $\tau_e = 0.125$ stands out as the landward-end is more distinctly muddy at greater distance from the sea, and muddy shoals are of smaller size, see Figure 11. Without further investigation the other cases seem to yield similar results, which can be attributed to the bed shear stresses in this domain not exceeding the respective erosion thresholds.

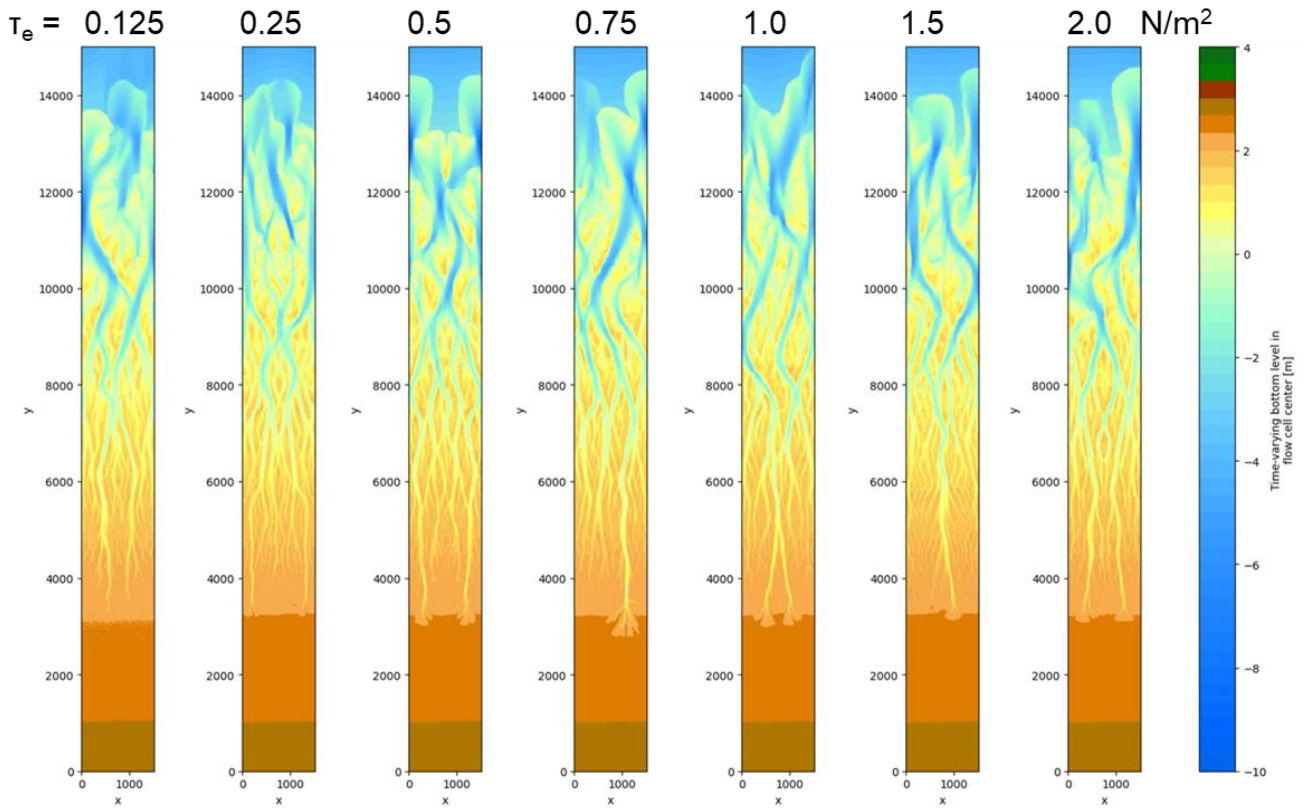


Figure 10 The bed level after 150 morphological years for $\tau_e = 0.125, 0.25, 0.5, 0.75, 1, 1.5, 2 \text{ N/m}^2$.

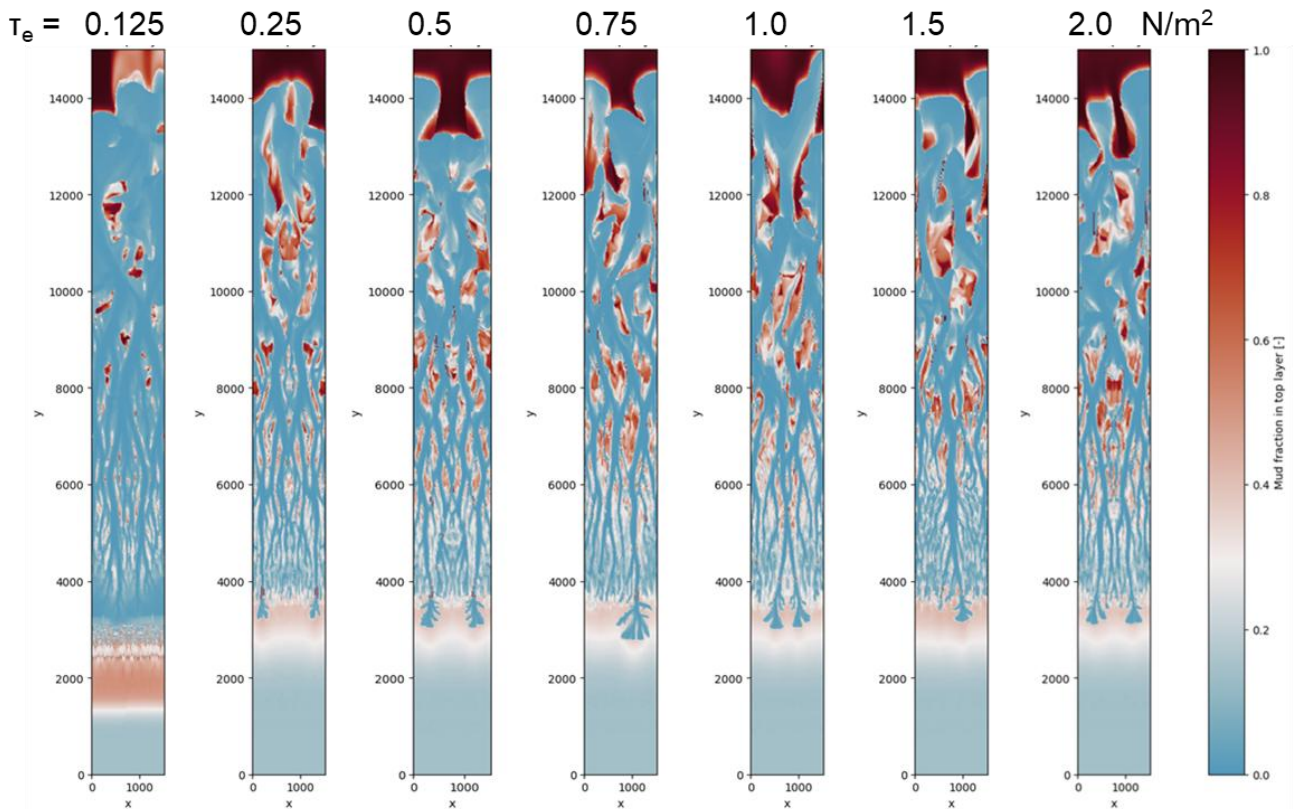


Figure 11 The mud fraction p_m in the active toplayer after 150 morphological years for $\tau_e = 0.125, 0.25, 0.5, 0.75, 1, 1.5, 2 \text{ N/m}^2$.

4 Next steps

Potential next steps in the model development are

- Advanced postprocessing (Strahler order, channel-flat ratio, etc following Colina-Alonso et al., 2023)
- Evaluating the effect of SLR
- Strengthen support for chosen sediment parameters
- Implementation of available functionalities (mangroves) and evaluating impact on channel development and adapting to SLR
- Implementing new functionalities in the model code (salt crust, algal mats) and evaluating impact

Prioritizing these next steps is part of the next project meeting.

5 References

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A Appendix

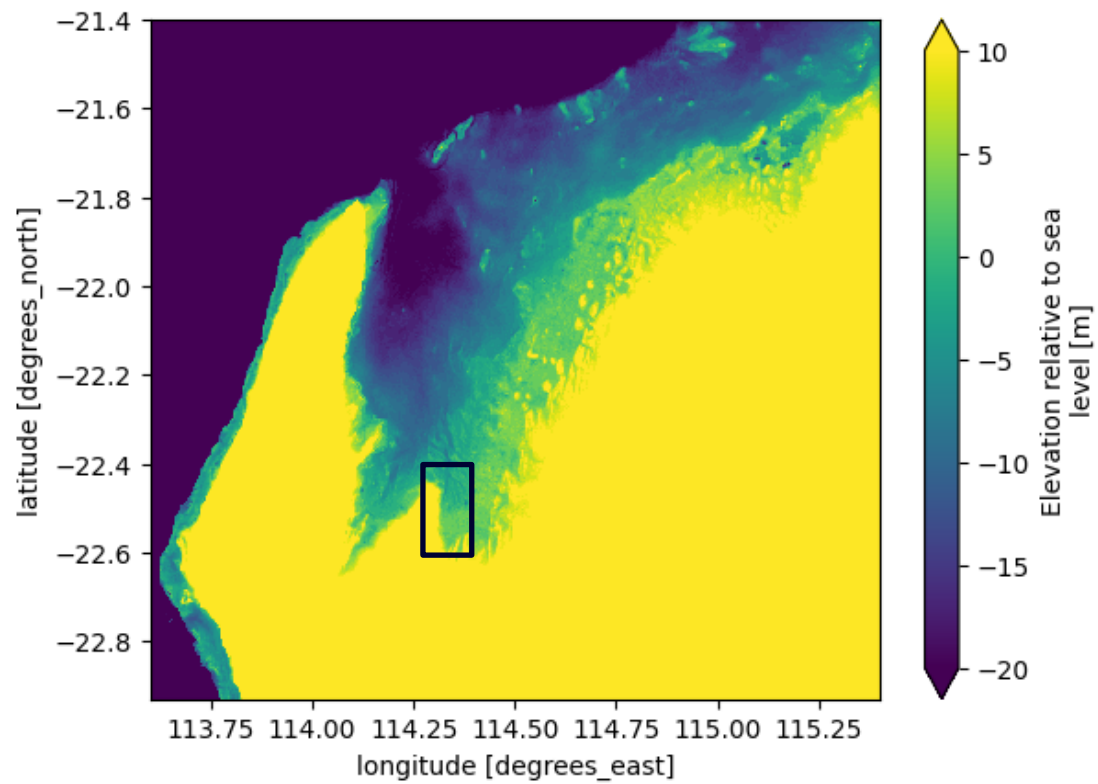


Figure 12 Bathymetry of Exmouth Bay, obtained from GEBCO (2023)

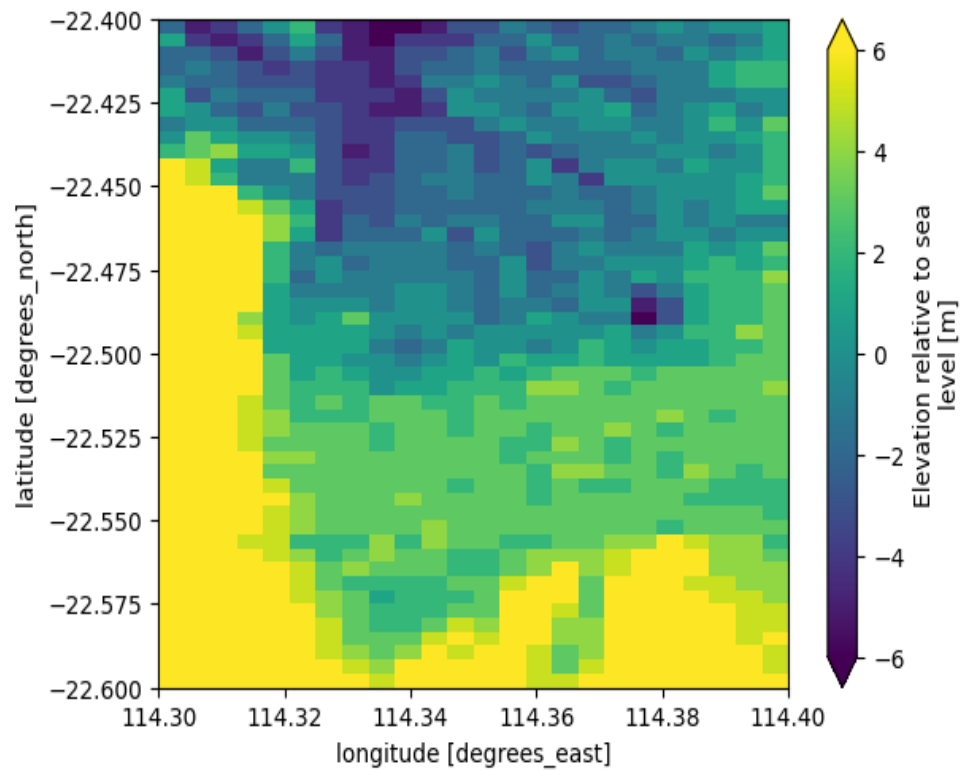


Figure 13 Bathymetry of the Giralía area, obtained from GEBCO (2023)